Challenges and fixes applied

This log summarizes the main product and implementation problems addressed in Study Desk. It describes the current code and database state; it is not a test report.

Applied fixes

Challenge	Fix applied	Current result
Re-uploading an identical PDF/DOCX repeated text extraction.	Added a `source_sha256` field and index in migration 5. The upload endpoint checks the raw file hash before parsing.	Byte-identical files are recognized early. If bytes differ, extracted-text hash detection still handles duplicates.
Re-indexing unchanged documents did unnecessary work.	Added content/chunker-version checks, chunk reuse, and reuse of existing embeddings; embeddings are processed in batches.	Unchanged indexed content can reuse saved chunks and completed embeddings.
Search and generated study materials needed page-level grounding.	Store extracted pages and indexed chunks with page ranges; return filenames, excerpts, and page references from retrieval.	Search, answers, quiz items, guide sections, and concept nodes can link back to source pages.
Answers could refer to a source label that was not supplied or a page outside the retrieved range.	Validate answer citation IDs and page ranges and return a citation warning when a citation is not verifiable.	The user can inspect source links and see a warning when the answer's inline citation needs review.
Quiz questions were generic, uneven in format, and could expose internal IDs such as S3.	Added type-specific prompts, distinct-question guidance, selectable type/difficulty/count, source-label cleanup, and a distributed passage sample. Larger quizzes are generated in batches of up to 10.	Learners choose multiple choice, short answer, or fill in the blank and 10–50 questions. Visible source links use filename/page labels.
Quiz review stopped at showing whether a choice was right.	Added hints, explanations, result counts, and a review path for missed questions.	A learner can retry only the questions they missed. Short answers remain self-assessed.
Study controls were crowded together.	Grouped flashcards, quizzes, study guides, and concept maps in separate responsive cards; kept quiz count visible in the quiz card; shortened the library hero.	Common study actions are separated by purpose and adapt to narrower screens.
A folder feature was confusing and overlapped with collections.	Removed folder controls from the active interface; retained collections as the supported way to group documents.	Collections are visible and supported. Legacy folder schema from an earlier migration may remain in the database.
Document deletion needed clearer consequences and a recovery path.	Added confirmation, names affected source files, backs up the database before deletion, removes explicit FTS/index/study references, and promotes a surviving duplicate when needed.	Deleting one document does not remove a shared source file still used by another library record.
A user needed a way to move or restore the whole library.	Added a ZIP backup containing the database and uploads, validation of restore contents, and a safety backup before replacement.	Backup/restore is available from the Library view.
Local AI or embeddings may be unavailable.	Added a sidebar status indicator and keyword-search fallback.	The app identifies unavailable Qwen/embedding models and can still provide keyword retrieval.
OCR added a separate dependency and setup burden.	Removed OCR integration, status UI, Tesseract configuration, and setup instructions at the user's request.	Image-only PDFs are unsupported; selectable text PDFs and DOCX remain supported.

Database migration history

The current copied project database is at migration version 5. Each migration script creates and verifies a timestamped database backup before changing the schema.

Version	Purpose
1	Multi-document study foundation, including collections, duplicate tracking, chunk/index tables, processing records, and supporting schema.
2	Persist generated flashcard and quiz sets.
3	Permit saved study guides.
4	Permit saved concept maps.
5	Store source-file hashes to recognize byte-identical re-uploads before text extraction.

Migration scripts are in `app/migrate_database*.py`. For a fresh database, follow the ordered commands in the root `README.md`. Do not skip a version.

Current limitations and follow-up notes

- The app does not perform OCR. Image-only PDFs will fail extraction and cannot be indexed.
- Short-answer grading is self-assessment; fill-in-the-blank uses normalized string comparison and can reject a valid synonym.
- 50-question quizzes require multiple local model calls and can take longer than smaller quizzes.
- The GET route for `/api/documents/{document_id}/file` is duplicated in `app/main.py`. The earlier handler validates the stored filename; a later duplicate handler should be removed so the route is declared once.
- No automated test suite is present in the current project tree. The migration scripts perform their own backup and database-integrity checks, but application behavior still needs manual review when changes are made.